Journal of Machine Learning Research 0 (2026) 1-10 Submitted Submitted 2/2026; Published Under Review

A Framework for Rigorous Evaluation of Model-Agnostic Explainability Methods: Multi-Metric Statistical Benchmarking, Operational Protocol, and Reproducibility

Jonathan Herrera-Vasquez JONNABIO@GMAIL.COM
UNADE

Editor: TBD

Abstract

Evaluating explainability methods requires more than a single faithfulness proxy. We present a modular benchmarking framework centered on quantitative XAI quality metrics: fidelity, stability, sparsity, computational cost, and faithfulness gap, plus an explicit method for operating the framework end-to-end. On the UCI Adult benchmark, we use a staged protocol (EXP1 calibration/reproducibility and EXP2 comparative/robustness benchmarking); the robustness cohort currently contains 250 of 300 planned configurations (83.3% coverage). Across complete model-size blocks (5 models, $N \in \{50, 100, 200\}$), Friedman tests indicate significant method differences for fidelity ($\chi^2 = 42.12$, $p = 3.78 \times 10^{-9}$), stability ($\chi^2 = 43.88$, $p = 1.60 \times 10^{-9}$), sparsity ($\chi^2 = 35.64$, $p = 8.92 \times 10^{-8}$), faithfulness gap ($\chi^2 = 45.00$, $p = 9.25 \times 10^{-10}$), and runtime ($\chi^2 = 27.72$, $p = 4.16 \times 10^{-6}$). SHAP leads on fidelity/stability, DiCE leads on sparsity, and LIME is generally fastest and most practical outside SVM-KernelSHAP bottlenecks. We release the framework, operation protocol, and artifacts with explicit data-quality caveats for reproducible benchmark use. LLM-based semantic scoring is deferred to a dedicated follow-up study.

Keywords: Explainable AI, benchmark methodology, reproducibility, statistical evaluation, model-agnostic explanations

1 Introduction

Evaluation of post-hoc explainability methods remains methodologically fragmented. Recent review evidence shows that XAI studies use heterogeneous metrics and often report inconsistent evaluation objectives across domains, which limits comparability and cumulative progress (Kadir et al., 2023; Pawlicki et al., 2024). In practice, fidelity-centric reporting is still common, even when stability, complexity, and computational burden materially affect whether explanations are usable in decision pipelines.

Open-source tooling has improved reproducibility at the implementation layer, including interfaces for large sets of explainers and metrics (Li et al., 2022; Hedström et al., 2023). However, tool-level coverage does not by itself establish inferential comparability: benchmark claims still require explicit controls for split discipline, artifact qualification, aggregation units, and multiplicity-aware statistical testing.

Parallel work in graph explainability benchmarking reinforces this point. GraphXAI, ShapeGGen, and B-XAIC show that benchmark value depends on explicit dataset assump-

©2026 Jonathan Herrera-Vasquez.

License: CC-BY 4.0, see https://creativecommons.org/licenses/by/4.0/. Attribution requirements are provided at http://jmlr.org/papers/v0/PAPER-A-0001.html.

tions, ground-truth design choices, and transparent reporting of evaluation constraints (Agarwal et al., 2023; Proszewska et al., 2025). At the same time, recent findings on attribution consensus in Rashomon sets indicate that explanation variability can persist even under similar predictive performance (Laberge et al., 2023). Faithfulness-oriented studies also show that metric behavior is sensitive to perturbation design and protocol choices, requiring robustness-aware interpretation (Zheng et al., 2025).

This paper focuses on model-agnostic, tabular post-hoc explanation benchmarking. It does not target representation-level interpretability (e.g., neuron-level analysis in NLP models), which has distinct constructs and evaluation methodologies (Durrani et al., 2023).

Given these gaps, we operationalize a benchmark-first framework that combines multi-metric scoring with an auditable operation protocol:

1. A generic trainer/explainer architecture for model-agnostic evaluation.
2. A prescriptive operation method that defines how the framework is run, audited, and reported.
3. Multi-metric quantitative scoring with explicit trade-off analysis.
4. Reproducibility and statistical rigor suitable for benchmark publication.

2 Related Work and Framework Design

2.1 Literature-Grounded Rationale

The literature motivating this design supports five requirements:

1. Multi-metric evaluation is necessary.
2. Faithfulness is multi-faceted.
3. Computational tractability must be reported explicitly.
4. Benchmark transparency and artifact accounting are first-order requirements.
5. LLM-judge scoring should be treated as a separate calibrated track.

2.2 Architecture

The framework decouples model training, explanation generation, and evaluation through a registry/factory design. Current model coverage includes `logreg`, `rf`, `xgb`, `svm`, and `mlp`, with `lime`, `shap`, `anchors`, and `dice` explainers.

2.3 Metric Suite (Primary)

Primary quantitative metrics are:

- Fidelity
- Stability

Table 1: Evidence cohorts and their role in Paper A claims.

Cohort	Primary objective	Role in claims
EXP1 pilot/calibration	Implementation sanity checks before scaled runs.	Not used for confirmatory significance claims.
EXP1 reproducibility	Repeated-seed variance profiling on core settings.	Supports reproducibility characterization (dispersion/CV).
EXP2 comparative	Broad method-model coverage under fixed seeds.	Descriptive comparative context.
EXP2 robustness	Multi-seed, multi-sample-size benchmark runs.	Primary inferential cohort for global and paired tests.

- Sparsity
- Cost
- Faithfulness gap

2.4 Paper A Operational Outcome: Framework Operation Method (FOM-7)

Paper A contributes not only benchmark findings, but also a reusable operating method for end-to-end benchmark execution, auditing, and claim formation.

3 Methodology

3.1 Study Scope and Evidence Cohorts

To separate development from confirmation, Paper A uses staged evidence cohorts with explicit claim roles.

3.2 Research Questions, Factors, and Analysis Units

The confirmatory questions are:

1. **RQ1 (global separation):** do explainer methods differ on the primary metrics under matched contexts?
2. **RQ2 (targeted pairwise contrast):** on matched cells, does SHAP differ from LIME on quality-cost trade-offs?
3. **RQ3 (reproducibility profile):** how much metric dispersion appears under repeated runs?

EXP2 is a crossed design with model family g, explainer method k, seed s, and per-quadrant sampling intensity n:

$$g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}, \quad k \in \{\text{shap}, \text{lime}, \text{anchors}, \text{dice}\},$$

$$s \in \{42, 123, 456, 789, 999\}, \quad n \in \{50, 100, 200\}.$$

Planned EXP2 robustness size is $5 \times 4 \times 5 \times 3 = 300$ runs. In the current snapshot, 250 artifacts are present and 233 are analyzable (16 empty, 1 malformed).

The inferential unit is not the instance row. For metric m, the run-level estimand is $y_{g,k,s,n,m}$ and the matched block is $b = (g, n)$. Tests are performed on run/block aggregates to avoid pseudo-replication.

3.3 Data Protocol and Leakage Controls

The benchmark uses Adult Income loaded through `src/data_loading/adult.py` with OpenML as primary source and UCI fallback. Cleaning applies whitespace normalization, missing-value canonicalization, target normalization, duplicate removal, and deterministic feature ordering.

Leakage controls are fixed across all explainers:

1. stratified split before any preprocessing fit,
2. train-only preprocessor fitting,
3. shared persisted preprocessor across methods in each run,
4. explanation on the same transformed feature space used by the model.

Data split is

```
train_test_split(test_size=0.2, stratify=y, random_state=seed).
```

For seed 42, the split is 39,032 train and 9,758 test rows. Each run samples TP/TN/FP/FN strata with target N per stratum (nominal run size $4N$, subject to stratum capacity).

3.4 Model and Explanation Protocol

EXP2 evaluates frozen EXP1 model artifacts; models are not retrained during EXP2. Model families are `logreg`, `rf`, `xgb`, `svm`, and `mlp`, with fixed repository-tracked settings.

All methods are executed through `ExplainerWrapper.explain_instance` and timed per instance. Effective explainer conditions are:

1. SHAP: tree mode for RF and XGB; kernel mode for LogReg, SVM, and MLP; background sample count set to 50.
2. LIME: `num_samples=1000`, `num_features=10`, `kernel_width=3.0`.
3. Anchors: `AnchorTabular` on transformed training data with effective threshold fixed at 0.95.
4. DiCE: random backend, opposite-class counterfactual target, importance from absolute original–counterfactual deltas.

Two implementation-bound caveats are explicit in claim interpretation: Anchors YAML threshold is not consumed at runtime (fixed 0.95), and DiCE YAML counterfactual count is not consumed (effective `total_CFs=1`).

3.5 Metrics and Statistical Inference

Primary metrics are fidelity, stability, sparsity, faithfulness gap, and cost (milliseconds). Aggregation is instance → run → block. EEU is not used for confirmatory inference in this draft.

Faithfulness and stability are computed as:

$$\Delta_k = |f(x) - f(x_{\text{mask-top-}k})|, \quad \rho = \operatorname{corr}(|w_i|, |f(x) - f(x_{-i})|), \tag{1}$$

$$x^{(t)} = x + \epsilon^{(t)},\ \epsilon^{(t)} \sim \mathcal{N}(0, \sigma^2), \quad S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right). \tag{2}$$

Inferential workflow:

1. Friedman omnibus test per metric on complete (g, n) blocks.
2. Nemenyi post-hoc localization when Friedman is significant.
3. Two-sided paired Wilcoxon SHAP–LIME tests on matched (g, s, n) cells (45-cell primary set; 65-cell sensitivity set).

Multiplicity handling uses Nemenyi within each metric and Holm-Bonferroni across the five primary metrics for each inferential family (Friedman, Wilcoxon-45, Wilcoxon-65). Reported uncertainty/effect summaries include mean, standard deviation, CV, confidence intervals (where generated), Kendall's W, Cohen's d_z, and median paired differences.

Artifact qualification is deterministic: malformed or empty runs are excluded, no synthetic reconstruction is used, and omnibus tests are restricted to block-complete contexts.

3.6 Reproducibility, FOM-7, and Validity Boundaries

Paper A operationalizes the benchmark using FOM-7, a stage-gated protocol:

1. protocol freeze,
2. controlled batch execution,
3. artifact integrity audit,
4. harmonization to analysis-ready tables,
5. inferential export,
6. reproducibility profiling,
7. claim-ready reporting.

Progression requires gate satisfaction at each stage; failed gates downgrade affected outputs to descriptive-only status.

The reproducibility package includes code, experiment configurations, model and preprocessor artifacts, per-run outputs, analysis exports, and figure scripts. For concision, command-level rerun instructions, provenance hashes, and environment anchors are maintained in repository manifests and scripts rather than duplicated in the main text.

Validity controls and residual risks are summarized as follows:

1. **Internal validity:** explicit missingness audit and block-complete inference; residual risk from non-random coverage gaps.
2. **Construct validity:** metric definitions tied to code-level implementations; residual risk from omitted secondary metrics in claims.
3. **Statistical conclusion validity:** matched-design non-parametric tests with multiplicity control; residual risk from absent equivalence bands.
4. **External validity:** diversity across model families and seeds, but single tabular dataset scope.
5. **Implementation validity:** effective runtime parameter bindings are disclosed; harmonization changes may shift absolute levels.

Scope of claims: conclusions in Section 4 are comparative and bounded to this benchmark setting (Adult tabular task, declared models/explainers, and artifact-qualified EXP2 subsets).

4 Results

4.1 Overview of Evaluation Setup

This section reports results from two evidence streams: EXP2 robustness (global and paired method comparisons) and EXP1 reproducibility (variance profiling). Compared explainers are SHAP, LIME, Anchors, and DiCE. Reported metrics are Fidelity, Stability, Sparsity, Faithfulness Gap, and Cost (ms). Metric definitions and the inferential workflow are specified in Section 3.5. "15 complete model-size blocks" denotes the block-level EXP2 units used for global inference, and "45 matched configs" denotes the SHAP–LIME subset with shared $(\text{model}, \text{seed}, N)$ cells on `logreg/rf/xgb`.

4.2 Global Trade-offs Across Methods (Quality vs Cost)

Table 2: Global method means (EXP2 robustness: 15 complete model-size blocks).

Method	Fidelity	Stability	Sparsity	Faithfulness Gap	Time (ms)
SHAP	0.8176	0.7377	0.2264	0.4474	685220.23
LIME	0.5602	0.0144	0.0846	0.3342	3660.68
Anchors	0.3853	0.0006	0.0928	0.2382	25326.92
DiCE	0.1716	0.3602	0.0164	0.0988	16306.50

SHAP has the highest Fidelity (0.8176), highest Stability (0.7377), and highest Faithfulness Gap (0.4474). DiCE has the lowest Sparsity value (0.0164), while LIME has the lowest Cost (3660.68 ms). Anchors shows very low Stability (0.0006) with Cost (25326.92 ms) above LIME and DiCE.

The reported means show an explicit quality–cost frontier: SHAP occupies the strongest quality region on Fidelity/Stability/Faithfulness Gap with the highest runtime (685220.23 ms), whereas LIME occupies the lowest-cost region with lower Fidelity and Stability. DiCE provides the lowest Sparsity value with lower Fidelity and Faithfulness Gap.

4.3 Global Statistical Evidence Across Methods

Table 3: Friedman tests across methods.

Metric	Statistic	p-value	Significant
Fidelity	42.12	3.78e-09	Yes
Stability	43.88	1.60e-09	Yes
Sparsity	35.64	8.92e-08	Yes
Faithfulness Gap	45.00	9.25e-10	Yes
Cost	27.72	4.16e-06	Yes

For every metric, the Friedman test rejects the null of equal performance across methods. This establishes global between-method differences, but does not identify pairwise direction without post-hoc localization. Post-hoc localization uses Nemenyi pairwise comparisons within each metric, with Holm-Bonferroni adjustment across the five primary metrics.

4.4 Targeted Pairwise Comparison (SHAP vs LIME on Shared Families)

Table 4: Paired Wilcoxon comparison on 45 matched configurations (`logreg/rf/xgb`).

Metric	SHAP Mean	LIME Mean	Wilcoxon p-value
Fidelity	0.8112	0.5556	5.68e-14
Stability	0.8013	0.0154	5.68e-14
Sparsity	0.3156	0.0845	5.68e-14
Faithfulness Gap	0.3924	0.3408	5.68e-14
Cost (ms)	1258.72	210.45	2.29e-06

Within this matched subset, SHAP is higher on Fidelity, Stability, and Faithfulness Gap, while LIME has lower Cost and lower Sparsity value. The reported p-values indicate non-zero paired differences for all listed metrics. Practically, the quality gains for SHAP are accompanied by a runtime penalty relative to LIME under shared cells.

This comparison is restricted to `logreg/rf/xgb` matched configurations and is not directly generalizable to non-matched families or to methods outside the SHAP–LIME pair.

4.5 Reproducibility and Variance Profile

EXP1 repeated-run evidence indicates lower dispersion for quality-oriented metrics and higher dispersion for runtime behavior. Quality metrics remain relatively stable across repeated seeds, while Cost varies more strongly for SHAP variants.

- Metric stability (quality): Fidelity CV ≤ 0.0634, Stability CV ≤ 0.0826, Sparsity CV ≤ 0.0441, Faithfulness-gap CV ≤ 0.0358.
- Runtime instability (cost): Cost CV reaches 0.225 for SHAP variants.

- Decision implication: quality rankings are more repeatable than runtime behavior under the reported repeated-run protocol.

4.6 Synthesis and Implications for Practice

- When the objective is maximizing Fidelity/Stability/Faithfulness Gap and compute budget is permissive, SHAP is the supported choice in this benchmark.
- When compute or latency is constrained, LIME is the supported low-cost option, with lower Fidelity and Stability than SHAP.
- When conciseness (Sparsity) is the primary objective, DiCE provides the lowest reported Sparsity value, with lower Fidelity and Faithfulness Gap.
- Methods are not statistically interchangeable: Friedman tests reject the null of equal performance across methods for all five primary metrics.
- On the shared SHAP–LIME subset, quality improvements for SHAP are paired with higher runtime cost.

Interpretation limits:

- Anchors stability values near zero should be interpreted with implementation caveats: effective runtime behavior uses fixed precision threshold 0.95 and binary rule-membership importance vectors, which can reduce perturbation-based similarity under this protocol.
- Method rankings are currently validated only on one tabular benchmark (Adult); external generalization requires replication on additional datasets and modalities before cross-domain claims are made.

5 Conclusion

The reported metrics indicate a multi-objective frontier rather than a single best method: SHAP is strongest on the reported quality-oriented metrics, LIME is strongest on runtime efficiency, and DiCE is strongest on the reported sparsity criterion. Global tests reject the null of equal method performance across the primary metrics, and paired SHAP–LIME analysis indicates a quality–cost trade-off on shared model families.

Methodologically, Paper A contributes FOM-7 as an auditable benchmark operation protocol that connects protocol freezing, controlled execution, integrity audit, deterministic inference export, and claim-traceable reporting. This contribution is operational and reproducibility-focused rather than algorithmic.

- **Takeaway:** method choice should be objective-driven (quality, sparsity, or runtime), not based on a single aggregated score.
- **Takeaway:** statistical evidence supports method differentiation at the omnibus level; pairwise direction must be interpreted within matched-design limits.
- **Limitation:** incomplete EXP2 coverage and malformed artifacts qualify confidence in full-grid frontier stability.

- **Next steps:** publish a missingness diagnostic by model/explainer/seed/sample-size and release a versioned reproducibility bundle with immutable DOI (code, configs, run outputs, inferential exports, and figure scripts).
- **Next steps:** integrate deferred semantic evaluation via a preregistered rubric, blinded human rating, inter-rater reliability reporting, and agreement analysis against the quantitative metrics.

Acknowledgments and Disclosure of Funding

Funding and competing-interest disclosures are to be finalized in the submission package. This draft was prepared from repository artifacts dated February 2026.

References

Chirag Agarwal, Owen Queen, Himabindu Lakkaraju, and Marinka Zitnik. 2023. Evaluating explainability for graph neural networks. *Scientific Data*, 10:144.

Nadir Durrani, Fahim Dalvi, and Hassan Sajjad. 2023. Discovering Salient Neurons in deep NLP models. *Journal of Machine Learning Research*, 24:1–40.

Anna Hedström, Leander Weber, Dilyara Bareeva, Daniel Krakowczyk, Franz Motzkus, Wojciech Samek, Sebastian Lapuschkin, and Marina M.-C. Höhne. 2023. Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond. *Journal of Machine Learning Research*, 24:1–11.

Md Abdul Kadir, Amir Mosavi, and Daniel Sonntag. 2023. Evaluation Metrics for XAI: A Review, Taxonomy, and Practical Applications. In *Proceedings of the 2023 IEEE 27th International Conference on Intelligent Engineering Systems (INES)*, pages 111–124, Nairobi, Kenya.

Gabriel Laberge, Yann Pequignot, Alexandre Mathieu, Foutse Khomh, and Mario Marchand. 2023. Partial Order in Chaos: Consensus on Feature Attributions in the Rashomon Set. *Journal of Machine Learning Research*, 24:1–50.

Xuhong Li, Haoyi Xiong, Xingjian Li, Xuanyu Wu, Zeyu Chen, and Dejing Dou. 2022. InterpretDL: Explaining Deep Models in PaddlePaddle. *Journal of Machine Learning Research*, 23:1–6.

Marek Pawlicki, Aleksandra Pawlicka, Federica Uccello, Sebastian Szelest, Salvatore D'Antonio, Rafał Kozik, and Michał Choraś. 2024. Evaluating the necessity of the multiple metrics for assessing explainable AI: A critical examination. *Neurocomputing*, 602:128282.

Magdalena Proszewska, Tomasz Danel, and Dawid Rymarczyk. 2025. B-XAIC Dataset: Benchmarking Explainable AI for Graph Neural Networks Using Chemical Data. *arXiv preprint arXiv:2505.22252*.

Xu Zheng, Farhad Shirani, Zhuomin Chen, Chaohao Lin, Wei Cheng, Wenbo Guo, and Dongsheng Luo. 2025. F-FIDELITY: A Robust Framework for Faithfulness Evaluation of Explainable AI. In *International Conference on Learning Representations (ICLR)*.